

Rockchip Developer Guide Linux GMAC Mode Configuration

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Preface

Overview

This document provides configuration examples for different modes of the Ethernet GMAC interface on the Rockchip platform, which is used to resolve Ethernet configuration issues.

Product Version

Chipset	Kernel Version
ROCKCHIP Chip	All versions

Intended Audience

This document (this guide) is mainly intended for:

Technical support engineers

Software development engineers

Revision History

Version	Author	Date	Change Description
V1.0.0	Wu Dachao	2021-01-26	Initial version
V1.1.0	Wu Dachao	2021-12-28	Support RK3588
V1.1.1	Wu Dachao	2022-11-28	Correct errors
V1.2.0	Wu Dachao	2022-11-29	Support RK3528
V1.3.0	Wu Dachao	2023-01-16	Support RK3562
V1.4.0	Wu Dachao	2024-04-26	Support RK3576, configuration reference RK3588

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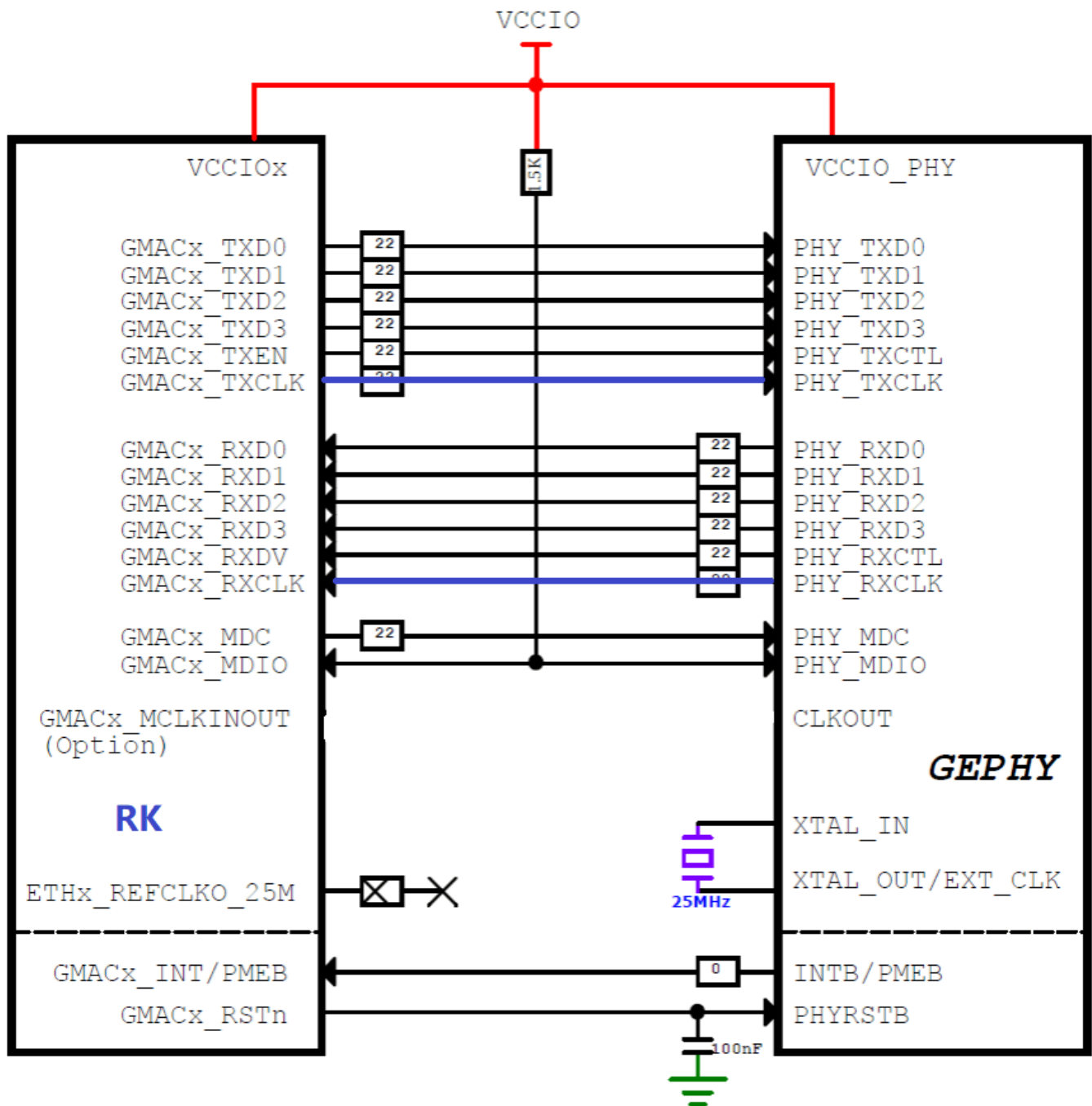
The main mode connection diagrams for RGMII/RMII are listed as follows:

1. RGMII Mode

When using the master PLL output clock method, the 125M clock provided by the PHY as an input method serves as an alternative solution.

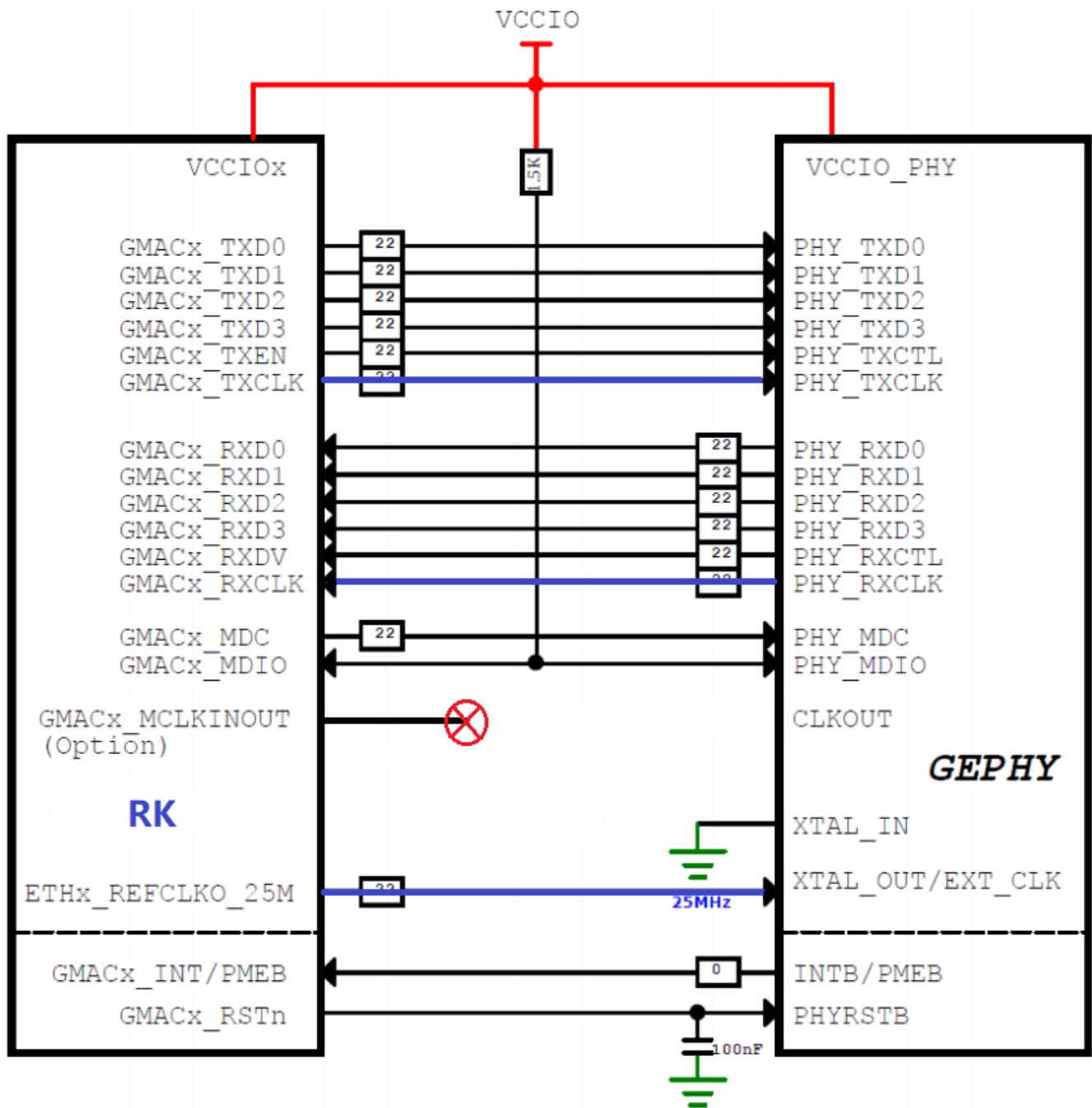
1.1 PLL Output 125M for TX_CLK, Crystal 25M for PHY

The main controller PLL provides the clock required for TXCLK, while the PHY 25M clock is supplied by the crystal oscillator.



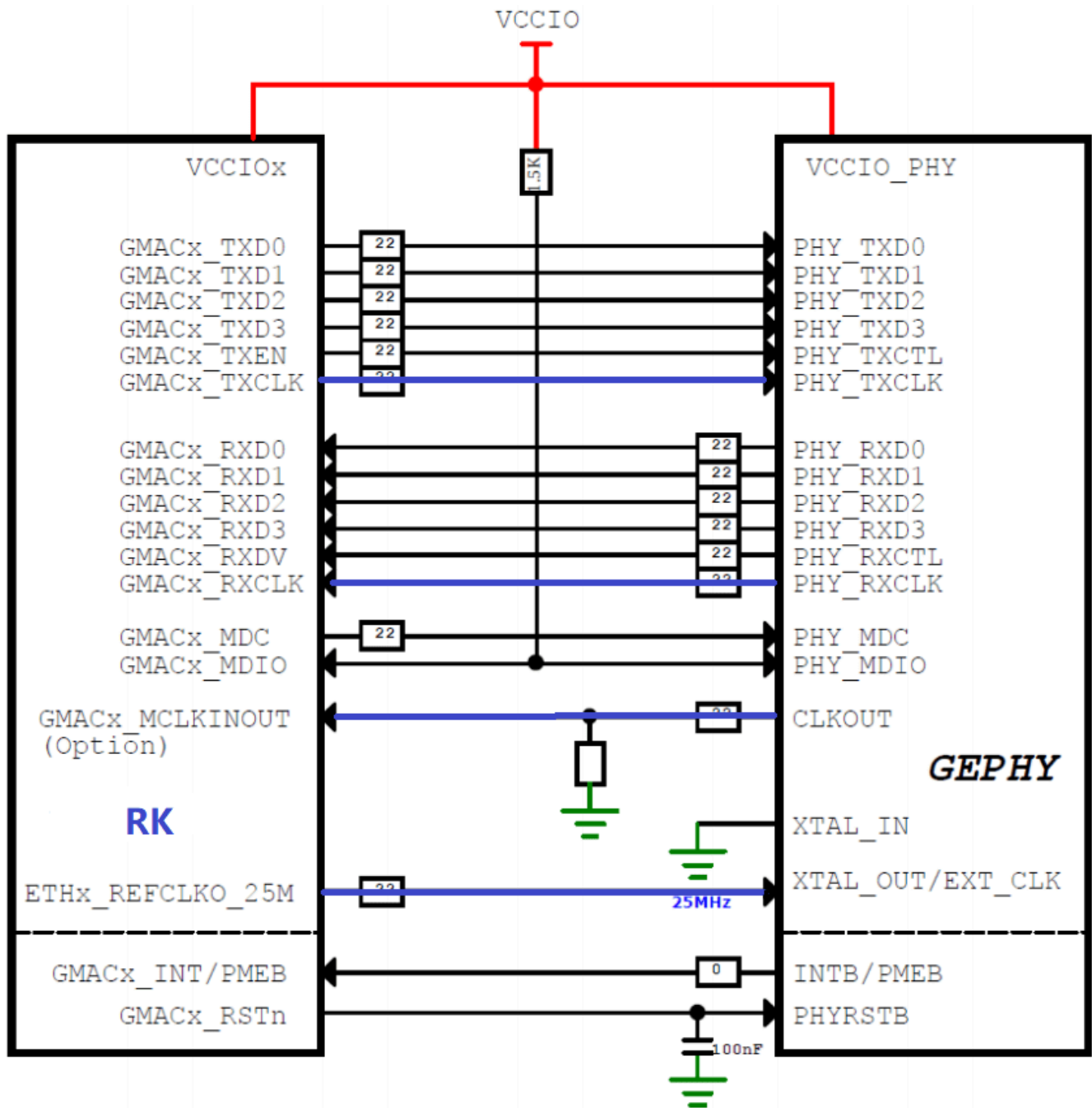
1.2 PLL Output 125M for TX_CLK, PLL 25M for PHY

The master PLL provides the clock required for TXCLK, and the PHY 25M clock is supplied by the master controller.



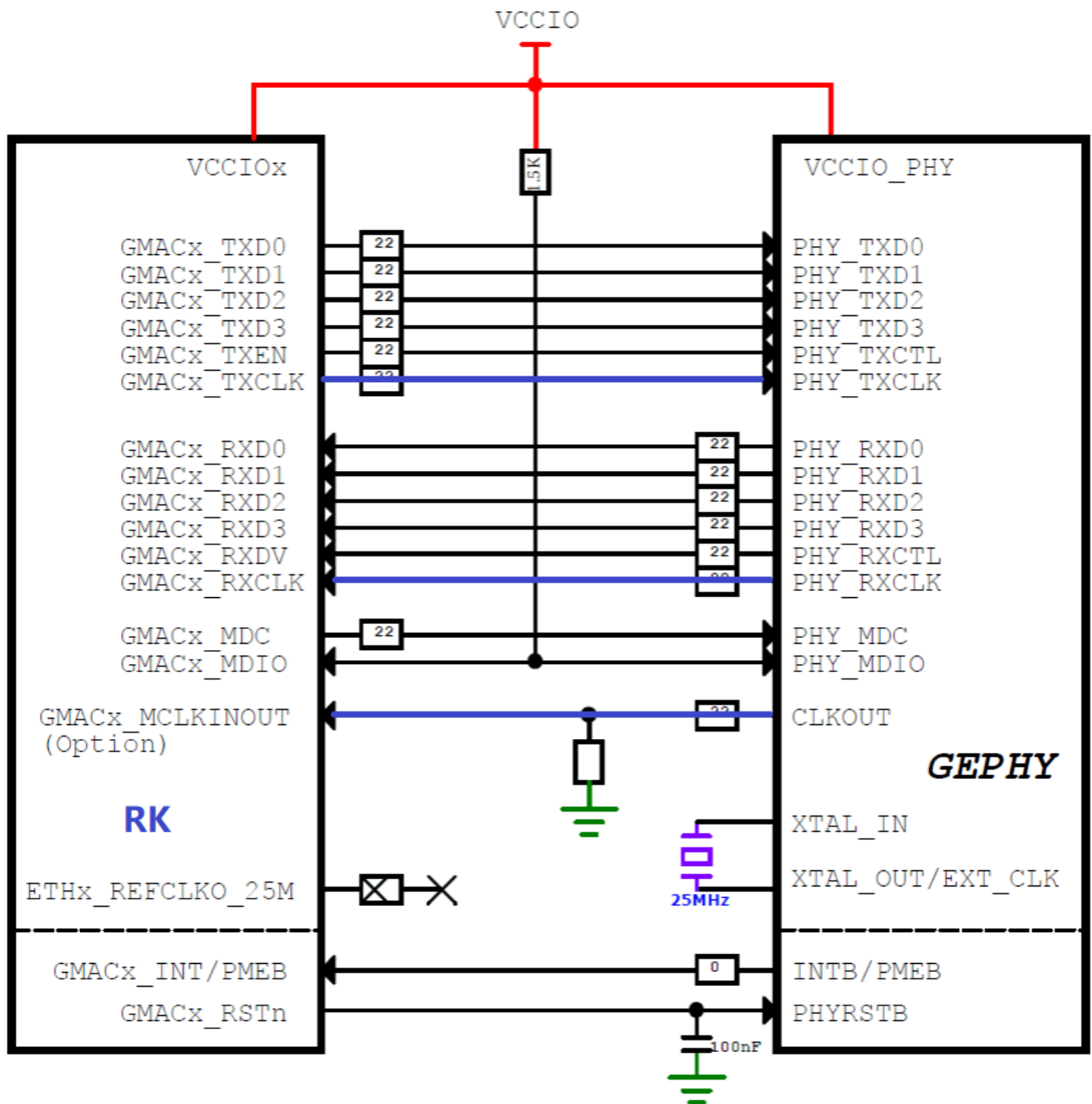
1.3 125M TX_CLK Input from PHY, PLL 25M for PHY

The required clock for TXCLK is provided by the PHY, and the 25M clock for the PHY is provided by the master controller.



1.4 125M TX_CLK input from PHY, Crystal 25M for PHY

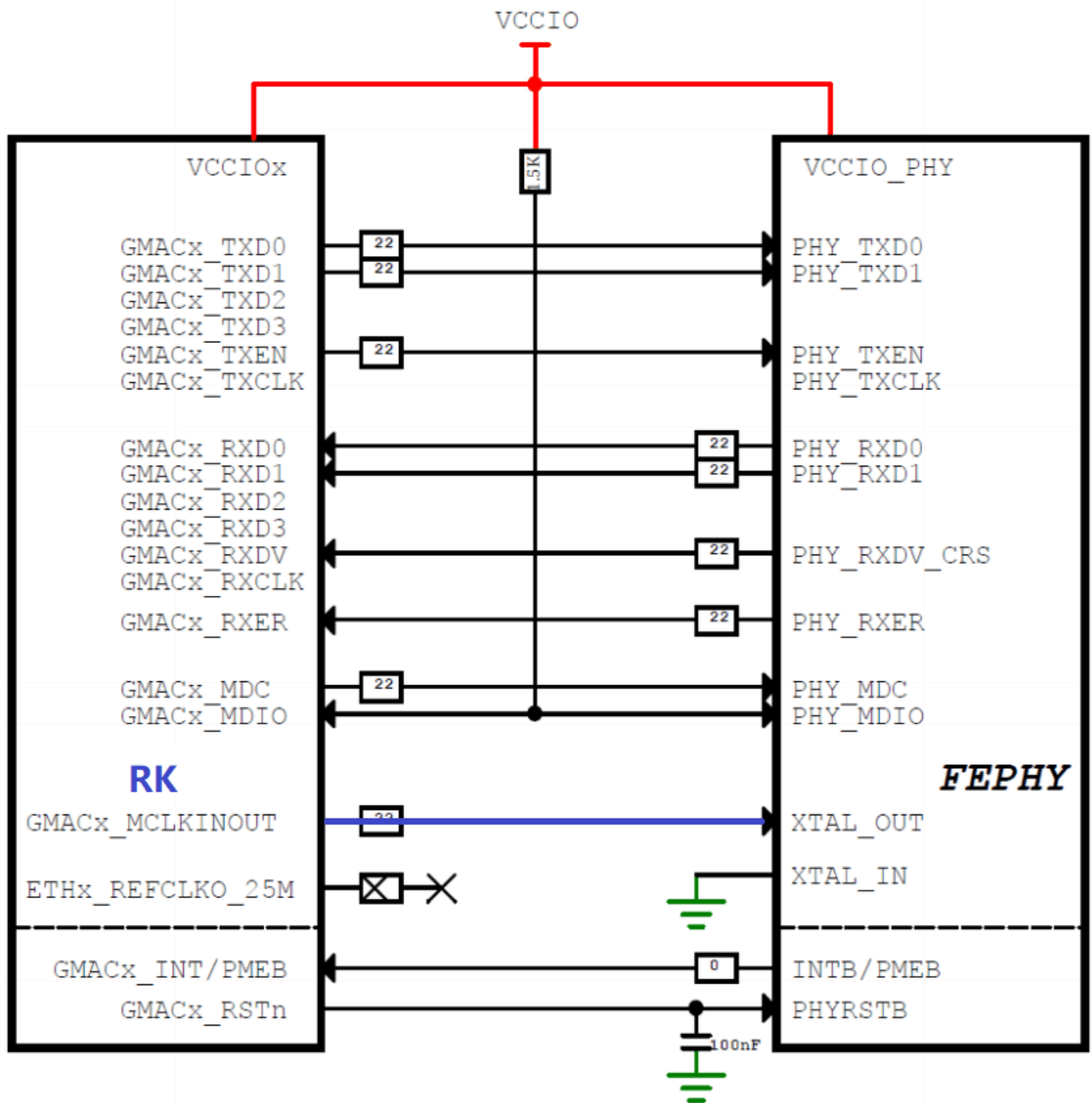
The TXCLK required clock is provided by the PHY, and the PHY 25M clock is supplied by the crystal oscillator.



2. RMII Mode

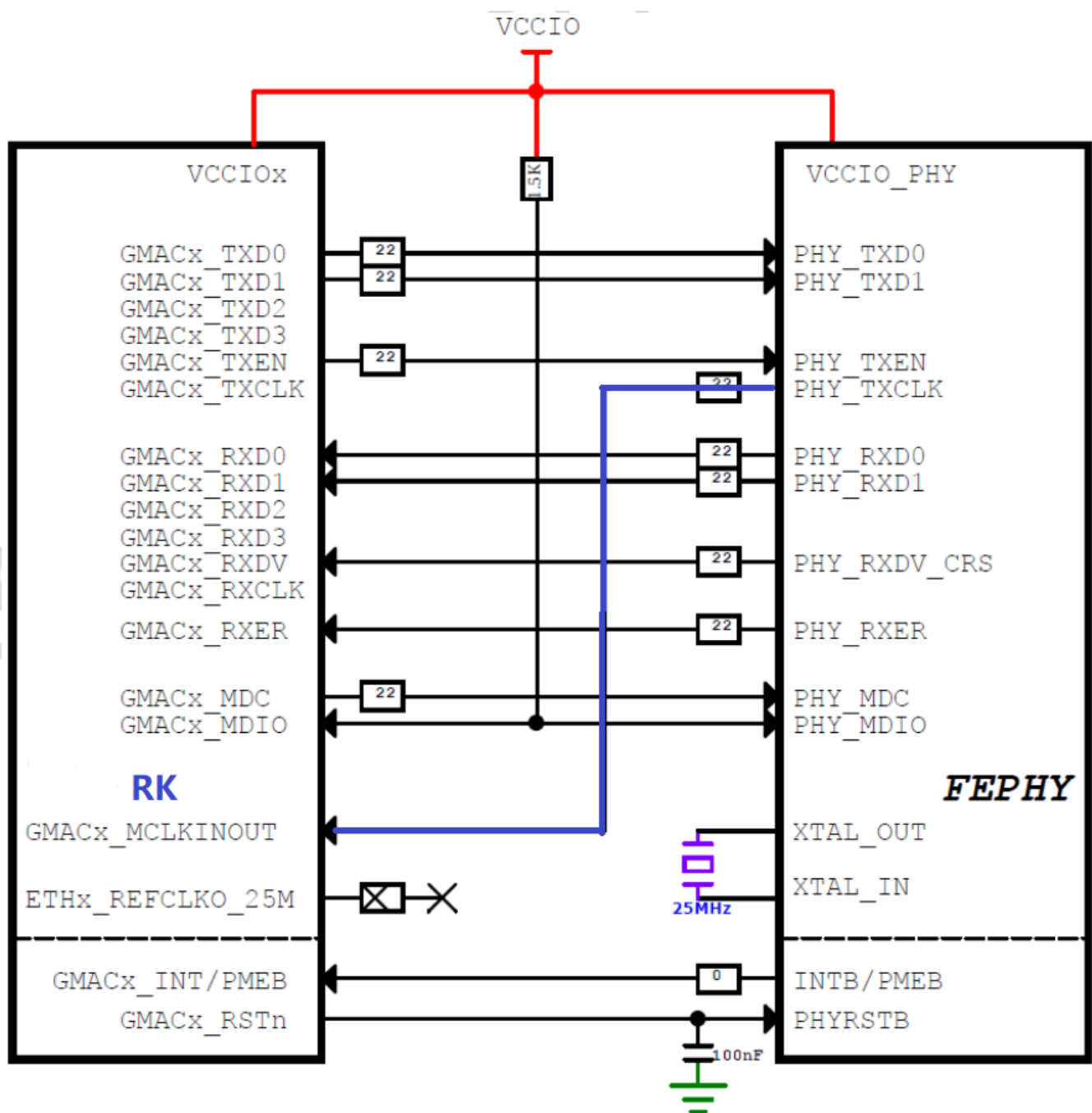
2.1 RMII Clock Output

The controller provides the required clock for RMII



2.2 RMII Clock Input

PHY provides the clock required for RMII



Similarly, in RGMII mode, the crystal can also be replaced by the master control output of 25M.

3. Mode Configuration

Configurations under different modes primarily include settings for phy mode, clock, and pinctrl, which are interrelated and need to be configured simultaneously; otherwise, they will not function properly. Below are references for different configuration methods under various chips, using the SDK board-level DTS as an example, with a focus on modifications to the '+' part within the gmac node in the dts.

3.1 PX30

3.1.1 PX30 RMII Clock Output

```
&gmac {
    phy-supply = <&vcc_phy>;
+    clock_in_out = "output";
+    assigned-clocks = <&cru SCLK_MAC>;
+    assigned-clock-rates = <50000000>;
    snps,reset-gpio = <&gpio2 13 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 50000 50000>;
+    pinctrl-names = "default";
+    pinctrl-0 = <&rmii_pins &mac_refclk_12ma>;
    status = "okay";
};
```

3.1.2 PX30 RMII Clock Input

```
&gmac {
    phy-supply = <&vcc_phy>;
+    clock_in_out = "input";
+    assigned-clocks = <&cru SCLK_MAC>;
+    assigned-clock-parents = <&gmac_clkin>;
    snps,reset-gpio = <&gpio2 13 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 50000 50000>;
+    pinctrl-names = "default";
+    pinctrl-0 = <&rmii_pins &mac_refclk>;
    status = "okay";
};
```

3.2 RK1808

3.2.1 RK1808 RMII Clock Output

```
&gmac {
    phy-supply = <&vcc_phy>;
+    phy-mode = "rmii";
+    clocks = <&cru SCLK_GMAC>, <&cru SCLK_GMAC_RX_TX>,
+            <&cru SCLK_GMAC_RX_TX>, <&cru SCLK_GMAC_REF>,
+            <&cru SCLK_GMAC_REFOUT>, <&cru ACLK_GMAC>,
+            <&cru PCLK_GMAC>, <&cru SCLK_GMAC_RMII_SPEED>;
+    clock-names = "stmmaceth", "mac_clk_rx",
+            "mac_clk_tx", "clk_mac_ref",
```

```

+             "clk_mac_refout", "aclk_mac",
+             "pclk_mac", "clk_mac_speed";
+     assigned-clocks = <&cru SCLK_GMAC_RX_TX>;
+     assigned-clock-parents = <&cru SCLK_GMAC_RMII_SPEED>;
snps,reset-gpio = <&gpio0 10 GPIO_ACTIVE_LOW>;
snps,reset-active-low;
snps,reset-delays-us = <0 50000 50000>;
+     pinctrl-names = "default";
+     pinctrl-0 = <&rmii_pins>;
status = "okay";
};

```

3.2.2 RK1808 RMII Clock Input

```

+&gmac_clkin {
+     clock-frequency = <50000000>;
+};

&gmac {
    phy-supply = <&vcc_phy>;
    phy-mode = "rmii";
+     clock_in_out = "input";
+     clocks = <&cru SCLK_GMAC>, <&cru SCLK_GMAC_RX_TX>,
+             <&cru SCLK_GMAC_RX_TX>, <&cru SCLK_GMAC_REF>,
+             <&cru SCLK_GMAC_REFOUT>, <&cru ACLK_GMAC>,
+             <&cru PCLK_GMAC>, <&cru SCLK_GMAC_RMII_SPEED>;
+     clock-names = "stmmaceth", "mac_clk_rx",
+                 "mac_clk_tx", "clk_mac_ref",
+                 "clk_mac_refout", "aclk_mac",
+                 "pclk_mac", "clk_mac_speed";
+     assigned-clocks = <&cru SCLK_GMAC_RX_TX>, <&cru SCLK_GMAC>;
+     assigned-clock-parents = <&cru SCLK_GMAC_RMII_SPEED>, <&gmac_clkin>;
snps,reset-gpio = <&gpio0 10 GPIO_ACTIVE_LOW>;
snps,reset-active-low;
snps,reset-delays-us = <0 50000 50000>;
+     pinctrl-names = "default";
+     pinctrl-0 = <&rmii_pins>;
status = "okay";
};

```

3.2.3 RK1808 RGMII Clock Output

```

&gmac {
    phy-supply = <&vcc_phy>;
+     phy-mode = "rgmii";
+     clock_in_out = "output";
+     assigned-clocks = <&cru SCLK_MAC>;
+     assigned-clock-rates = <125000000>;
snps,reset-gpio = <&gpio0 10 GPIO_ACTIVE_LOW>;

```

```

snps,reset-active-low;
/* Reset time is 20ms, 100ms for rtl8211f */
snps,reset-delays-us = <0 20000 100000>;
tx_delay = <0x50>;
rx_delay = <0x3a>;
status = "okay";
};

```

3.2.4 RK1808 RGMII Clock Input

```

&gmac {
    phy-supply = <&vcc_phy>;
+    phy-mode = "rgmii";
+    clock_in_out = "input";
+    assigned-clocks = <&cru SCLK_GMAC>;
+    assigned-clock-parents = <&gmac_clkin>;
    snps,reset-gpio = <&gpio0 10 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;
    tx_delay = <0x50>;
    rx_delay = <0x3a>;
    status = "okay";
};

```

3.3 RK3128

3.3.1 RK3128 RMII Clock Output

```

&gmac {
+    assigned-clocks = <&cru SCLK_MAC_SRC>;
+    assigned-clock-rates = <50000000>;
+    clock_in_out = "output";
+    pinctrl-names = "default";
+    pinctrl-0 = <&rmii_pins>;
    phy-supply = <&vcc_phy>;
+    phy-mode = "rmii";
    snps,reset-active-low;
    snps,reset-delays-us = <0 10000 50000>;
    snps,reset-gpio = <&gpio2 24 GPIO_ACTIVE_LOW>;
    status = "okay";
};

```

3.3.2 RK3128 RMI Clock Input

```
+&clkin_gmac {
+    clock-frequency = <50000000>;
+};

&gmac {
+    assigned-clocks = <&cru SCLK_MAC>;
+    assigned-clock-parents = <&clkin_gmac>;
+    clock_in_out = "input";
+    pinctrl-names = "default";
+    pinctrl-0 = <&rmii_pins>;
+    phy-supply = <&vcc_phy>;
+    phy-mode = "rmii";
+    snps,reset-active-low;
+    snps,reset-delays-us = <0 10000 50000>;
+    snps,reset-gpio = <&gpio2 24 GPIO_ACTIVE_LOW>;
+    status = "okay";
};
```

3.3.3 RK3128 RGMII Clock Input

```
&gmac {
+    assigned-clocks = <&cru SCLK_MAC>;
+    assigned-clock-parents = <&clkin_gmac>;
+    clock_in_out = "input";
+    pinctrl-names = "default";
+    pinctrl-0 = <&rgmii_pins>;
+    phy-supply = <&vcc_phy>;
+    phy-mode = "rgmii";
+    snps,reset-active-low;
+    snps,reset-delays-us = <0 20000 100000>;
+    snps,reset-gpio = <&gpio2 24 GPIO_ACTIVE_LOW>;
+    tx_delay = <0x30>;
+    rx_delay = <0x16>;
+    status = "okay";
};
```

3.4 RK3228

3.4.1 RK3228 RMI Clock Output

```
&gmac {
+    assigned-clocks = <&cru SCLK_MAC_EXTCLK>, <&cru SCLK_MAC>;
+    assigned-clock-parents = <&ext_gmac>, <&cru SCLK_MAC_EXTCLK>;
+    assigned-clock-rates = <0>, <50000000>;
```

```

+     clock_in_out = "output";
    phy-supply = <&vcc_phy>;
+     phy-mode = "rmii";
+     pinctrl-names = "default";
+     pinctrl-0 = <&rmii_pins>;
    snps,reset-gpio = <&gpio2 RK_PD0 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 20000 100000>;
    status = "okay";
};

```

3.4.2 RK3228 RMII Clock Input

```

+&ext_gmac: external-gmac-clock {
+     clock-frequency = <50000000>;
+}

&gmac {
+     assigned-clocks = <&cru SCLK_MAC_EXTCLK>, <&cru SCLK_MAC>;
+     assigned-clock-parents = <&ext_gmac>, <&cru SCLK_MAC_EXTCLK>;
+     clock_in_out = "input";
    phy-supply = <&vcc_phy>;
+     phy-mode = "rmii";
+     pinctrl-names = "default";
+     pinctrl-0 = <&rmii_pins>;
    snps,reset-gpio = <&gpio2 RK_PD0 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 20000 100000>;
    status = "okay";
};

```

3.4.3 RK3228 RGMII Clock Output

```

&gmac {
+     assigned-clocks = <&cru SCLK_MAC_EXTCLK>, <&cru SCLK_MAC>;
+     assigned-clock-parents = <&ext_gmac>, <&cru SCLK_MAC_EXTCLK>;
+     assigned-clock-rates = <0>, <125000000>;
+     clock_in_out = "output";
    phy-supply = <&vcc_phy>;
+     phy-mode = "rgmii";
+     pinctrl-names = "default";
+     pinctrl-0 = <&rgmii_pins>;
    snps,reset-gpio = <&gpio2 RK_PD0 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 20000 100000>;
    tx_delay = <0x30>;
    rx_delay = <0x10>;
    status = "okay";
};

```

```
};
```

3.4.4 RK3228 RGMII Clock Input

```
&gmac {
+   assigned-clocks = <&cru SCLK_MAC_EXTCLK>, <&cru SCLK_MAC>;
+   assigned-clock-parents = <&ext_gmac>, <&cru SCLK_MAC_EXTCLK>;
+   clock_in_out = "input";
  phy-supply = <&vcc_phy>;
+   phy-mode = "rgmii";
+   pinctrl-names = "default";
+   pinctrl-0 = <&rgmii_pins>;
  snps,reset-gpio = <&gpio2 RK_PD0 GPIO_ACTIVE_LOW>;
  snps,reset-active-low;
  snps,reset-delays-us = <0 20000 100000>;
  tx_delay = <0x30>;
  rx_delay = <0x10>;
  status = "okay";
};
```

3.4.5 RK3228 Internal EPHY

```
&gmac {
+   assigned-clocks = <&cru SCLK_MAC_SRC>;
+   assigned-clock-rates = <50000000>;
+   clock_in_out = "output";
  phy-supply = <&vcc_phy>;
+   phy-mode = "RMII";
  phy-handle = <&phy>;
  status = "okay";

  mdio {
    compatible = "snps,dwmac-mdio";
    #address-cells = <1>;
    #size-cells = <0>;

    phy: ethernet-phy@0 {
      compatible = "ethernet-phy-id1234.d400", "ethernet-phy-ieee802.3-
c22";

      reg = <0>;
      clocks = <&cru SCLK_MAC_PHY>;
      resets = <&cru SRST_MACPHY>;
      phy-is-integrated;
    };
  };
};
```

3.5 RK3288

3.5.1 RK3288 RMI Clock Output

```
&gmac {
    phy-supply = <&vcc_phy>;
+    phy-mode = "rmii";
+    clock_in_out = "output";
+    assigned-clocks = <&cru SCLK_MAC>;
+    assigned-clock-rates = <50000000>;
    snps,reset-gpio = <&gpio4 RK_PA7 GPIO_ACTIVE_HIGH>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 20000 1000000>;
+    pinctrl-names = "default";
+    pinctrl-0 = <&rmii_pins>;
    status = "okay";
};
```

3.5.2 RK3288 RMI Clock Input

```
+&ext_gmac: external-gmac-clock {
+    clock-frequency = <50000000>;
+}

&gmac {
    phy-supply = <&vcc_phy>;
+    phy-mode = "rmii";
+    clock_in_out = "input";
+    assigned-clocks = <&cru SCLK_MAC>;
+    assigned-clock-parents = <&ext_gmac>;
    snps,reset-gpio = <&gpio4 RK_PA7 GPIO_ACTIVE_HIGH>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 20000 1000000>;
+    pinctrl-names = "default";
+    pinctrl-0 = <&rmii_pins>;
    status = "okay";
};
```

3.5.3 RK3288 RGMII Clock Input

```
&gmac {
    phy-supply = <&vcc_phy>;
+    phy-mode = "rgmii";
+    clock_in_out = "input";
    snps,reset-gpio = <&gpio4 RK_PA7 GPIO_ACTIVE_HIGH>;
    snps,reset-active-low;
```

```

        snps,reset-delays-us = <0 20000 1000000>;
+       assigned-clocks = <&cru SCLK_MAC>;
+       assigned-clock-parents = <&ext_gmac>;
+       pinctrl-names = "default";
+       pinctrl-0 = <&rgmii_pins>;
        tx_delay = <0x30>;
        rx_delay = <0x10>;
        status = "okay";
};

```

3.6 RK3328

3.6.1 RK3328 RMII Clock Output

```

&gmac2io {
    phy-supply = <&vcc_phy>;
+    phy-mode = "rmii";
+    clock_in_out = "output";
+    assigned-clocks = <&cru SCLK_MAC2IO>;
+    assigned-clock-rates = <50000000>;
    snps,reset-gpio = <&gpio1 RK_PC2 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 20000 100000>;
+    pinctrl-names = "default";
+    pinctrl-0 = <&rmiim1_pins>;
    status = "okay";
};

```

3.6.2 RK3328 RMII Clock Input

```

+&clkin_gmac {
+    clock-frequency = <50000000>;
+};

&gmac2io {
    phy-supply = <&vcc_phy>;
+    phy-mode = "rmii";
+    clock_in_out = "input";
+    assigned-clocks = <&cru SCLK_MAC2IO>, <&cru SCLK_MAC2IO_EXT>;
+    assigned-clock-parents = <&gmac_clkin>, <&gmac_clkin>;
    snps,reset-gpio = <&gpio1 RK_PC2 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 20000 100000>;
+    pinctrl-names = "default";
+    pinctrl-0 = <&rmiim1_pins>;
    status = "okay";
};

```

3.6.3 RK3328 RGMII Clock Input

```
&gmac2io {
    phy-supply = <&vcc_phy>;
+    phy-mode = "rgmii";
+    clock_in_out = "input";
+    assigned-clocks = <&cru SCLK_MAC2IO>, <&cru SCLK_MAC2IO_EXT>;
+    assigned-clock-parents = <&gmac_clkin>, <&gmac_clkin>;
    snps,reset-gpio = <&gpio1 RK_PC2 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 20000 100000>;
+    pinctrl-names = "default";
+    pinctrl-0 = <&rgmiim1_pins>;
    tx_delay = <0x26>;
    rx_delay = <0x11>;
    status = "okay";
};
```

3.6.4 RK3328 Internal EPHY

```
&gmac2phy {
    phy-supply = <&vcc_phy>;
+    clock_in_out = "output";
+    assigned-clocks = <&cru SCLK_MAC2PHY_SRC>;
+    assigned-clock-rate = <50000000>;
+    assigned-clocks = <&cru SCLK_MAC2PHY>;
+    assigned-clock-parents = <&cru SCLK_MAC2PHY_SRC>;
    status = "okay";
};
```

3.7 RK3368

3.7.1 RK3368 RMII Clock Output

```
&gmac {
    phy-supply = <&vcc_lan>;
+    phy-mode = "rmii";
+    clock_in_out = "output";
+    assigned-clocks = <&cru SCLK_MAC>;
+    assigned-clock-rates = <50000000>;
    snps,reset-gpio = <&gpio3 12 0>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 20000 100000>;
+    pinctrl-names = "default";
+    pinctrl-0 = <&rmii_pins>;
    status = "ok";
};
```

```
};
```

3.7.2 RK3368 RMII Clock Input

```
+&ext_gmac {
+    clock-frequency = <50000000>;
+}

&gmac {
    phy-supply = <&vcc_lan>;
+    phy-mode = "rmii";
+    clock_in_out = "input";
+    assigned-clocks = <&cru SCLK_MAC>;
+    assigned-clock-parents = <&ext_gmac>;
    snps,reset-gpio = <&gpio3 12 0>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 20000 100000>;
+    pinctrl-names = "default";
+    pinctrl-0 = <&rmii_pins>;
    status = "ok";
};
```

3.7.3 RK3368 RGMII Clock Input

```
&gmac {
    phy-supply = <&vcc_lan>;
+    phy-mode = "rmii";
+    clock_in_out = "input";
+    assigned-clocks = <&cru SCLK_MAC>;
+    assigned-clock-parents = <&ext_gmac>;
    snps,reset-gpio = <&gpio3 12 0>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 20000 100000>;
+    pinctrl-names = "default";
+    pinctrl-0 = <&rmii_pins>;
    status = "okay";
};
```

3.8 RK3399

3.8.1 RK3399 RMII Clock Output

```
&gmac {
+    assigned-clocks = <&cru SCLK_MAC>;
+    assigned-clock-rates = <50000000>;
```

```

+     clock_in_out = "output";
phy-supply = <&vcc_phy>;
+     phy-mode = "rmii";
+     pinctrl-names = "default";
+     pinctrl-0 = <&rmii_pins>;
snps,reset-gpio = <&gpio3 RK_PB7 GPIO_ACTIVE_LOW>;
snps,reset-active-low;
snps,reset-delays-us = <0 20000 100000>;
status = "okay";
};

```

3.8.2 RK3399 RMII Clock Input

```

+&clkkin_gmac {
+     clock-frequency = <50000000>;
+};

&gmac {
+     assigned-clocks = <&cru SCLK_RMII_SRC>;
+     assigned-clock-parents = <&clkkin_gmac>;
+     clock_in_out = "input";
phy-supply = <&vcc_phy>;
+     phy-mode = "rmii";
+     pinctrl-names = "default";
+     pinctrl-0 = <&rmii_pins>;
snps,reset-gpio = <&gpio3 RK_PB7 GPIO_ACTIVE_LOW>;
snps,reset-active-low;
snps,reset-delays-us = <0 20000 100000>;
status = "okay";
};

```

3.8.3 RK3399 RGMII Clock Input

```

&gmac {
+     assigned-clocks = <&cru SCLK_RMII_SRC>;
+     assigned-clock-parents = <&clkkin_gmac>;
+     clock_in_out = "input";
phy-supply = <&vcc_phy>;
+     phy-mode = "rgmii";
+     pinctrl-names = "default";
+     pinctrl-0 = <&rgmii_pins>;
snps,reset-gpio = <&gpio3 RK_PB7 GPIO_ACTIVE_LOW>;
snps,reset-active-low;
snps,reset-delays-us = <0 20000 100000>;
tx_delay = <0x28>;
rx_delay = <0x11>;
status = "okay";
};

```

3.9 RK3528

3.9.1 GMAC1 RMII Clock 50M Output, PLL 25M Output

```
&gmac1 {
    phy-mode = "rmii";
    clock_in_out = "output";

    snps,reset-gpio = <&gpio3 RK_PC3 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 10000 50000>;

    pinctrl-names = "default";
    pinctrl-0 = <&rgmii_miim
        &rgmii_tx_bus2
        &rgmii_rx_bus2
        &rgmii_clk
        &eth_pins>;

    phy-handle = <&rmii1_phy>;
    status = "okay";
};

&mdio1 {
    rmii1_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
        clocks = <&cru CLK_GMAC1_VPU_25M>;
    };
};
```

3.9.2 GMAC1 RMII Clock 50M Input, PLL 25M Output

```
&gmac1 {
    phy-mode = "rmii";
    clock_in_out = "input";

    snps,reset-gpio = <&gpio3 RK_PC3 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 10000 50000>;

    pinctrl-names = "default";
    pinctrl-0 = <&rgmii_miim
        &rgmii_tx_bus2
        &rgmii_rx_bus2
        &rgmii_clk
        &eth_pins>;
```

```

    phy-handle = <&rmii1_phy>;
    status = "okay";
};

&mdio1 {
    rmii1_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
        clocks = <&cru CLK_GMAC1_VPU_25M>;
    };
};

```

3.9.3 GMAC1 RGMII PLL Output 25M for PHY, PLL Output 125M for TX_CLK

```

&gmac1 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rgmii-rxid";
    clock_in_out = "output";

    snps,reset-gpio = <&gpio4 RK_PC2 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    tx_delay = <0x30>;
    /* rx_delay = <0x3f>; */

    pinctrl-names = "default";
    pinctrl-0 = <&rgmii_miim
        &rgmii_tx_bus2
        &rgmii_rx_bus2
        &rgmii_rgmii_clk
        &rgmii_rgmii_bus
        &eth_pins>;

    phy-handle = <&rgmii_phy>;
    status = "okay";
};

&mdio1 {
    rgmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
        clocks = <&cru CLK_GMAC1_VPU_25M>;
    };
};

```

3.9.4 GMAC1 RGMII Crystal 25M for PHY, PLL output 125M for TX_CLK

```
&gmac1 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rgmii-rxid";
    clock_in_out = "output";

    snps,reset-gpio = <&gpio4 RK_PC2 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    tx_delay = <0x30>;
    /* rx_delay = <0x3f>; */

    pinctrl-names = "default";
    pinctrl-0 = <&rgmii_miim
                &rgmii_tx_bus2
                &rgmii_rx_bus2
                &rgmii_rgmii_clk
                &rgmii_rgmii_bus>;

    phy-handle = <&rgmii_phy>;
    status = "okay";
};

&mdio1 {
    rgmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
    };
};
```

3.9.5 GMAC1 RGMII PLL Output 25M for PHY, RGMII_CLK Input 125M for TX_CLK

```
&gmac1 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rgmii-rxid";
    clock_in_out = "input";

    snps,reset-gpio = <&gpio4 RK_PC2 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    tx_delay = <0x30>;
    /* rx_delay = <0x3f>; */

    pinctrl-names = "default";
    pinctrl-0 = <&rgmii_miim
```

```

        &rgmii_tx_bus2
        &rgmii_rx_bus2
        &rgmii_rgmii_clk
        &rgmii_rgmii_bus
        &rgmii_clk
        &eth_pins>;

    phy-handle = <&rgmii_phy>;
    status = "okay";
};

&mdio1 {
    rgmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
        clocks = <&cru CLK_GMAC1_VPU_25M>;
    };
};

```

3.9.6 GMAC1 RGMII Crystal 25M for PHY, RGMII_CLK input 125M for TX_CLK

```

&gmac1 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rgmii-rxid";
    clock_in_out = "input";

    snps,reset-gpio = <&gpio4 RK_PC2 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    tx_delay = <0x30>;
    /* rx_delay = <0x3f>; */

    pinctrl-names = "default";
    pinctrl-0 = <&rgmii_miim
        &rgmii_tx_bus2
        &rgmii_rx_bus2
        &rgmii_rgmii_clk
        &rgmii_rgmii_bus
        &rgmii_clk>;

    phy-handle = <&rgmii_phy>;
    status = "okay";
};

&mdio1 {
    rgmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
    };
};

```

```
};
```

3.9.7 GMAC0 & FEPHY

GMAC0 is connected internally with FEPHY and is fixed to RMI, with no configurable modes; however, the PHY LED function can be configured based on the hardware schematic. There are 3 functional IOs that can be configured by setting the corresponding IO iomux, with the default configuration as follows:

```
&rmii0_phy {
    pinctrl-names = "default";
    pinctrl-0 = <&fephy0_led_link &fephy0_led_spd>;
};
```

3.10 RK3562

3.10.1 RK3562 RMI Clock Output

- gmac0m0

```
&gmac0 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rmii";
    clock_in_out = "output";

    snps,reset-gpio = <&gpio4 RK_PB1 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    pinctrl-names = "default";
    pinctrl-0 = <&rgmiim0_miim
        &rgmiim0_tx_bus2
        &rgmiim0_rx_bus2
        &rgmiim0_clk>;

    phy-handle = <&rmii_phy>;
    status = "okay";
};

&mdio0 {
    rmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
    };
};
```

- gmac0m1:

```

&gmac0 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rmii";
    clock_in_out = "output";

    snps,reset-gpio = <&gpio0 RK_PB0 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    pinctrl-names = "default";
    pinctrl-0 = <&rgmiim1_miim
        &rgmiim1_tx_bus2
        &rgmiim1_rx_bus2
        &rgmiim1_clk>;

    phy-handle = <&rmii_phy>;
    status = "okay";
};

&mdio0 {
    rmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
    };
};

```

- gmac1(MAC100):

```

&gmac1 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rmii";
    clock_in_out = "output";

    snps,reset-gpio = <&gpio0 RK_PB0 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    pinctrl-names = "default";
    pinctrl-0 = <&rmii_pins>;

    phy-handle = <&rmii_phy>;
    status = "okay";
};

&mdio1 {
    rmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
    };
};

```

3.10.2 RMI Clock Input, PLL out 25M for PHY

- gmac0m0

```
&gmac0 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rmii";
    clock_in_out = "input";

    snps,reset-gpio = <&gpio4 RK_PB1 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    pinctrl-names = "default";
    pinctrl-0 = <&rgmiim0_miim
        &rgmiim0_tx_bus2
        &rgmiim0_rx_bus2
        &rgmiim0_clk
        &ethm0_pins>;

    phy-handle = <&rmii_phy>;
    status = "okay";
};

&mdio0 {
    rmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
        clocks = <&cru CLK_GMAC_ETH_OUT2IO>;
        assigned-clocks = <&cru CLK_GMAC_ETH_OUT2IO>;
        assigned-clock-rates = <25000000>;
    };
};
```

- gmac0m1:

```
&gmac0 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rmii";
    clock_in_out = "input";

    snps,reset-gpio = <&gpio0 RK_PB0 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    pinctrl-names = "default";
    pinctrl-0 = <&rgmiim1_miim
        &rgmiim1_tx_bus2
        &rgmiim1_rx_bus2
```

```

        &rgmiim1_clk
        &ethm1_pins>;

    phy-handle = <&rmii_phy>;
    status = "okay";
};

&mdio0 {
    rmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
        clocks = <&cru CLK_GMAC_ETH_OUT2IO>;
        assigned-clocks = <&cru CLK_GMAC_ETH_OUT2IO>;
        assigned-clock-rates = <25000000>;
    };
};

```

- gmac1(MAC100):

```

&gmac1 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rmii";
    clock_in_out = "input";

    snps,reset-gpio = <&gpio0 RK_PB0 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    pinctrl-names = "default";
    pinctrl-0 = <&rmii_pins
        &ethm1_pins>;

    phy-handle = <&rmii_phy>;
    status = "okay";
};

&mdio1 {
    rmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
        clocks = <&cru CLK_GMAC_ETH_OUT2IO>;
        assigned-clocks = <&cru CLK_GMAC_ETH_OUT2IO>;
        assigned-clock-rates = <25000000>;
    };
};

```

3.10.3 RGMII PLL Output 25M for PHY, PLL Output 125M for TX_CLK

- gmac0m0

```
&gmac0 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rgmii-rxid";
    clock_in_out = "output";

    snps,reset-gpio = <&gpio3 RK_PA0 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    tx_delay = <0x3f>;
    /* rx_delay = <0x3f>; */

    pinctrl-names = "default";
    pinctrl-0 = <&rgmiim0_miim
        &rgmiim0_tx_bus2
        &rgmiim0_rx_bus2
        &rgmiim0_rgmii_clk
        &rgmiim0_rgmii_bus
        &ethm0_pins>;

    phy-handle = <&rgmii_phy>;
    status = "okay";
};

&mdio0 {
    rgmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
        clocks = <&cru CLK_GMAC_ETH_OUT2IO>;
        assigned-clocks = <&cru CLK_GMAC_ETH_OUT2IO>;
        assigned-clock-rates = <25000000>;
    };
};
```

- gmac0m1

```
&gmac0 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rgmii-rxid";
    clock_in_out = "output";

    snps,reset-gpio = <&gpio0 RK_PB0 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;
```

```

tx_delay = <0x3f>;
/* rx_delay = <0x3f>; */

pinctrl-names = "default";
pinctrl-0 = <&rgmiim1_miim
            &rgmiim1_tx_bus2
            &rgmiim1_rx_bus2
            &rgmiim1_rgmii_clk
            &rgmiim1_rgmii_bus
            &ethm1_pins>;

phy-handle = <&rgmii_phy>;
status = "okay";
};

&mdio0 {
    rgmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
        clocks = <&cru CLK_GMAC_ETH_OUT2IO>;
        assigned-clocks = <&cru CLK_GMAC_ETH_OUT2IO>;
        assigned-clock-rates = <25000000>;
    };
};

```

3.10.4 RGMII PLL Output 25M for PHY, RGMII_CLK Input 125M for TX_CLK

- gmac0m0

```

&gmac0 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rgmii-rxid";
    clock_in_out = "input";

    snps,reset-gpio = <&gpio3 RK_PA0 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    tx_delay = <0x3f>;
    /* rx_delay = <0x3f>; */

    pinctrl-names = "default";
    pinctrl-0 = <&rgmiim0_miim
                &rgmiim0_tx_bus2
                &rgmiim0_rx_bus2
                &rgmiim0_rgmii_clk
                &rgmiim0_rgmii_bus
                &rgmiim0_clk
                &ethm0_pins>;

```

```

    phy-handle = <&rgmii_phy>;
    status = "okay";
};

&mdio0 {
    rgmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
        clocks = <&cru CLK_GMAC_ETH_OUT2IO>;
        assigned-clocks = <&cru CLK_GMAC_ETH_OUT2IO>;
        assigned-clock-rates = <25000000>;
    }
};

```

- gmac0m1

```

&gmac0 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rgmii-rxid";
    clock_in_out = "input";

    snps,reset-gpio = <&gpio0 RK_PB0 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    tx_delay = <0x39>;
    /* rx_delay = <0x3f>; */

    pinctrl-names = "default";
    pinctrl-0 = <&rgmiim1_miim
        &rgmiim1_tx_bus2
        &rgmiim1_rx_bus2
        &rgmiim1_rgmii_clk
        &rgmiim1_rgmii_bus
        &rgmiim1_clk
        &ethm1_pins>;

    phy-handle = <&rgmii_phy>;
    status = "okay";
};

&mdio0 {
    rgmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
        clocks = <&cru CLK_GMAC_ETH_OUT2IO>;
        assigned-clocks = <&cru CLK_GMAC_ETH_OUT2IO>;
        assigned-clock-rates = <25000000>;
    }
};

```

3.10.5 RGMII Crystal 25M for PHY, PLL output 125M for TX_CLK

- gmac0m0

```
&gmac0 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rgmii-rxid";
    clock_in_out = "output";

    snps,reset-gpio = <&gpio3 RK_PA0 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    tx_delay = <0x3f>;
    /* rx_delay = <0x3f>; */

    pinctrl-names = "default";
    pinctrl-0 = <&rgmiim0_miim
                &rgmiim0_tx_bus2
                &rgmiim0_rx_bus2
                &rgmiim0_rgmii_clk
                &rgmiim0_rgmii_bus>;

    phy-handle = <&rgmii_phy>;
    status = "okay";
};

&mdio0 {
    rgmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
    };
};
```

- gmac0m1

```
&gmac0 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rgmii-rxid";
    clock_in_out = "output";

    snps,reset-gpio = <&gpio0 RK_PB0 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    tx_delay = <0x3f>;
    /* rx_delay = <0x3f>; */

    pinctrl-names = "default";
```

```

pinctrl-0 = <&rgmiim1_miim
            &rgmiim1_tx_bus2
            &rgmiim1_rx_bus2
            &rgmiim1_rgmii_clk
            &rgmiim1_rgmii_bus>;

phy-handle = <&rgmii_phy>;
status = "okay";
};

&mdio0 {
    rgmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
    };
};

```

3.10.6 RGMII Crystal 25M for PHY, RGMII_CLK input 125M for TX_CLK

- gmac0m0

```

&gmac0 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rgmii-rxid";
    clock_in_out = "input";

    snps,reset-gpio = <&gpio3 RK_PA0 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    tx_delay = <0x3f>;
    /* rx_delay = <0x3f>; */

    pinctrl-names = "default";
    pinctrl-0 = <&rgmiim0_miim
                &rgmiim0_tx_bus2
                &rgmiim0_rx_bus2
                &rgmiim0_rgmii_clk
                &rgmiim0_rgmii_bus
                &rgmiim0_clk>;

    phy-handle = <&rgmii_phy>;
    status = "okay";
};

&mdio0 {
    rgmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
    };
};

```

```
};
```

- gmac0m1

```
&gmac0 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    phy-mode = "rgmii-rxid";
    clock_in_out = "input";

    snps,reset-gpio = <&gpio0 RK_PB0 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    tx_delay = <0x39>;
    /* rx_delay = <0x3f>; */

    pinctrl-names = "default";
    pinctrl-0 = <&rgmiim1_miim
        &rgmiim1_tx_bus2
        &rgmiim1_rx_bus2
        &rgmiim1_rgmii_clk
        &rgmiim1_rgmii_bus
        &rgmiim1_clk>;

    phy-handle = <&rgmii_phy>;
    status = "okay";
};

&mdio0 {
    rgmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
    };
};
```

3.11 RK3568

3.11.1 RK3568 RMII Clock Output

- gmac0

```
&gmac0 {
+     phy-mode = "rmii";
+     clock_in_out = "output";
+     assigned-clocks = <&cru SCLK_GMAC0_RX_TX>, <&cru SCLK_GMAC0>;
+     assigned-clock-parents = <&cru SCLK_GMAC0_RMII_SPEED>;
+     assigned-clock-rates = <0>, <500000000>;
```

```

snps,reset-gpio = <&gpio3 RK_PC2 GPIO_ACTIVE_LOW>;
snps,reset-active-low;
snps,reset-delays-us = <0 20000 100000>;
pinctrl-names = "default";
pinctrl-0 = <&gmac0_miim &gmac0_clkout &gmac0_rx_bus2 &gmac0_tx_bus2>;

phy-handle = <&rmii_phy0>;
status = "okay";

};

&mdio0 {
    rmii_phy0: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};

```

- gmac1m0:

```

&gmac1 {
+   phy-mode = "rmii";
+   clock_in_out = "output";
+   assigned-clocks = <&cru SCLK_GMAC1_RX_TX>, <&cru SCLK_GMAC1>;
+   assigned-clock-parents = <&cru SCLK_GMAC1_RMII_SPEED>;
+   assigned-clock-rates = <0>, <50000000>;

snps,reset-gpio = <&gpio3 RK_PC2 GPIO_ACTIVE_LOW>;
snps,reset-active-low;
snps,reset-delays-us = <0 20000 100000>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&gmac1m0_miim &gmac1m0_clkout &gmac1m0_rx_bus2 &gmac1m0_tx_bus2>;

phy-handle = <&rmii_phy1>;
status = "okay";
};

&mdio1 {
    rmii_phy1: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};

```

- gmac1m1:

```

&gmac1 {
+   phy-mode = "rmii";
+   clock_in_out = "output";
+   assigned-clocks = <&cru SCLK_GMAC1_RX_TX>, <&cru SCLK_GMAC1>;
+   assigned-clock-parents = <&cru SCLK_GMAC1_RMII_SPEED>;
+   assigned-clock-rates = <0>, <50000000>;

```

```

snps,reset-gpio = <&gpio3 RK_PC2 GPIO_ACTIVE_LOW>;
snps,reset-active-low;
snps,reset-delays-us = <0 20000 100000>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&gmac1m1_miim &gmac1m1_clkout &gmac1m1_rx_bus2 &gmac1m1_tx_bus2>;

phy-handle = <&rmii_phy1>;
status = "okay";
};

&mdio1 {
    rmii_phy1: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};

```

3.11.2 RK3568 RMII Clock Input

- gmac0

```

+&gmac0_clk{
+   clock-frequency = <50000000>;
+};

&gmac0 {
+   phy-mode = "rmii";
+   clock_in_out = "input";

snps,reset-gpio = <&gpio3 RK_PC2 GPIO_ACTIVE_LOW>;
snps,reset-active-low;
snps,reset-delays-us = <0 20000 100000>;

+   assigned-clocks = <&cru SCLK_GMAC0_RX_TX>, <&cru SCLK_GMAC0>;
+   assigned-clock-parents = <&cru SCLK_GMAC0_RMII_SPEED>, <&gmac0_clk>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&gmac0_miim &gmac0_clkout &gmac0_rx_bus2 &gmac0_tx_bus2>;

phy-handle = <&rmii_phy0>;
status = "okay";
};

&mdio0 {
    rmii_phy0: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};

```

- gmac1m0:

```
+&gmac1_clkkin{
+    clock-frequency = <50000000>;
+};

&gmac1 {
+    phy-mode = "rmii";
+    clock_in_out = "input";

    snps,reset-gpio = <&gpio3 RK_PC2 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 20000 100000>;
+    assigned-clocks = <&cru SCLK_GMAC1_RX_TX>, <&cru SCLK_GMAC1>;
+    assigned-clock-parents = <&cru SCLK_GMAC1_RMII_SPEED>, <&gmac1_clkkin>;

+    pinctrl-names = "default";
+    pinctrl-0 = <&gmac1m0_miim &gmac1m0_clkkinout &gmac1m0_rx_bus2 &gmac1m0_tx_bus2>;

    phy-handle = <&rmii_phy1>;
    status = "okay";
};

&mdio1 {
    rmii_phy1: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};
```

- gmac1m1:

```
+&gmac1_clkkin{
+    clock-frequency = <50000000>;
+};

&gmac1 {
+    phy-mode = "rmii";
+    clock_in_out = "input";

    snps,reset-gpio = <&gpio3 RK_PC2 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 20000 100000>;
+    assigned-clocks = <&cru SCLK_GMAC1_RX_TX>, <&cru SCLK_GMAC1>;
+    assigned-clock-parents = <&cru SCLK_GMAC1_RMII_SPEED>, <&gmac1_clkkin>;

+    pinctrl-names = "default";
+    pinctrl-0 = <&gmac1m1_miim &gmac1m1_clkkinout &gmac1m1_rx_bus2 &gmac1m1_tx_bus2>;

    phy-handle = <&rmii_phy1>;
    status = "okay";
};
```

```
&mdio1 {
    rmii_phy1: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};
```

3.11.3 RGMII PLL Output 25M for PHY, PLL Output 125M for TX_CLK

- gmac0

```
&gmac0 {
+   phy-mode = "rgmii";
+   clock_in_out = "output";
+   assigned-clocks = <&cru SCLK_GMAC0_RX_TX>, <&cru SCLK_GMAC0>, <&cru CLK_MAC0_OUT>;
+   assigned-clock-parents = <&cru SCLK_GMAC0_RGMII_SPEED>;
+   assigned-clock-rates = <0>, <125000000>, <25000000>;

    snps,reset-gpio = <&gpio2 RK_PD3 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&gmac0_miim
+               &gmac0_tx_bus2
+               &gmac0_rx_bus2
+               &gmac0_rgmii_clk
+               &gmac0_rgmii_bus
+               &eth0_pins>;

    tx_delay = <0x3c>;
    rx_delay = <0x2f>;
    phy-handle = <&rgmii_phy0>;
    status = "okay";
};

&mdio0 {
    rgmii_phy0: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
+       clocks = <&cru CLK_MAC0_OUT>;
    };
};
```

- gmac1m0

```
&gmac1 {
+   phy-mode = "rgmii";
+   clock_in_out = "output";
```

```

snps,reset-gpio = <&gpio2 RK_PD1 GPIO_ACTIVE_LOW>;
snps,reset-active-low;
/* Reset time is 20ms, 100ms for rtl8211f */
snps,reset-delays-us = <0 20000 100000>;

+   assigned-clocks = <&cru SCLK_GMAC1_RX_TX>, <&cru SCLK_GMAC1>, <&cru CLK_MAC1_OUT>;
+   assigned-clock-parents = <&cru SCLK_GMAC1_RGMII_SPEED>;
+   assigned-clock-rates = <0>, <125000000>, <25000000>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&gmac1m0_miim
+               &gmac1m0_tx_bus2
+               &gmac1m0_rx_bus2
+               &gmac1m0_rgmii_clk
+               &gmac1m0_rgmii_bus
+               &eth1m0_pins>;

tx_delay = <0x4f>;
rx_delay = <0x26>;

phy-handle = <&rgmii_phy1>;
status = "okay";
};

&mdio1 {
    rgmii_phy1: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
+       clocks = <&cru CLK_MAC1_OUT>;
    };
};

```

- gmac1m1

```

&gmac1 {
+   phy-mode = "rgmii";
+   clock_in_out = "output";

snps,reset-gpio = <&gpio2 RK_PD1 GPIO_ACTIVE_LOW>;
snps,reset-active-low;
/* Reset time is 20ms, 100ms for rtl8211f */
snps,reset-delays-us = <0 20000 100000>;

+   assigned-clocks = <&cru SCLK_GMAC1_RX_TX>, <&cru SCLK_GMAC1>, <&cru CLK_MAC1_OUT>;
+   assigned-clock-parents = <&cru SCLK_GMAC1_RGMII_SPEED>;
+   assigned-clock-rates = <0>, <125000000>, <25000000>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&gmac1m1_miim
+               &gmac1m1_tx_bus2
+               &gmac1m1_rx_bus2
+               &gmac1m1_rgmii_clk

```

```

+          &gmac1m1_rgmii_bus
+          &eth1m1_pins>;

    tx_delay = <0x4f>;
    rx_delay = <0x26>;

    phy-handle = <&rgmii_phy1>;
    status = "okay";
};

&mdio1 {
    rgmii_phy1: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
+        clocks = <&cru CLK_MAC1_OUT>;
    };
};

```

3.11.4 RGMII PLL Output 25M for PHY, RGMII_CLK Input 125M for TX_CLK

- gmac0

```

&gmac0 {
+    phy-mode = "rgmii";
+    clock_in_out = "input";
+    assigned-clocks = <&cru SCLK_GMAC0_RX_TX>, <&cru SCLK_GMAC0>, <&cru CLK_MAC0_OUT>;
+    assigned-clock-parents = <&cru SCLK_GMAC0_RGMII_SPEED>, <&gmac0_clkin>;
+    assigned-clock-rates = <0>, <125000000>, <250000000>;

    snps,reset-gpio = <&gpio2 RK_PD3 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

+    pinctrl-names = "default";
+    pinctrl-0 = <&gmac0_miim
+                &gmac0_tx_bus2
+                &gmac0_rx_bus2
+                &gmac0_rgmii_clk
+                &gmac0_rgmii_bus
+                &eth0_pins
+                &gmac0_clkinout>;

    tx_delay = <0x3c>;
    rx_delay = <0x2f>;
    phy-handle = <&rgmii_phy0>;
    status = "okay";
};

&mdio0 {
    rgmii_phy0: phy@0 {

```

```

        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
+        clocks = <&cru CLK_MAC0_OUT>;
    };
};

```

- gmac1m0

```

&gmac1 {
+    phy-mode = "rgmii";
+    clock_in_out = "input";

    snps,reset-gpio = <&gpio2 RK_PD1 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

+    assigned-clocks = <&cru SCLK_GMAC1_RX_TX>, <&cru SCLK_GMAC1>, <&cru CLK_MAC1_OUT>;
+    assigned-clock-parents = <&cru SCLK_GMAC1_RGMII_SPEED>, <&gmac1_clkin>;
+    assigned-clock-rates = <0>, <125000000>, <25000000>;

+    pinctrl-names = "default";
+    pinctrl-0 = <&gmac1m0_miim
+                &gmac1m0_tx_bus2
+                &gmac1m0_rx_bus2
+                &gmac1m0_rgmii_clk
+                &gmac1m0_rgmii_bus
+                &eth1m0_pins
+                &gmac1m0_clkinout>;

    tx_delay = <0x4f>;
    rx_delay = <0x26>;

    phy-handle = <&rgmii_phy1>;
    status = "okay";
};

&mdio0 {
    rgmii_phy1: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
+        clocks = <&cru CLK_MAC0_OUT>;
    };
};

```

- gmac1m1

```

&gmac1 {
+    phy-mode = "rgmii";
+    clock_in_out = "input";

    snps,reset-gpio = <&gpio2 RK_PD1 GPIO_ACTIVE_LOW>;

```

```

snps,reset-active-low;
/* Reset time is 20ms, 100ms for rtl8211f */
snps,reset-delays-us = <0 20000 100000>;

+   assigned-clocks = <&cru SCLK_GMAC1_RX_TX>, <&cru SCLK_GMAC1>, <&cru CLK_MAC1_OUT>;
+   assigned-clock-parents = <&cru SCLK_GMAC1_RGMII_SPEED>, <&gmac1_clkin>;
+   assigned-clock-rates = <0>, <125000000>, <25000000>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&gmac1m1_miim
+               &gmac1m1_tx_bus2
+               &gmac1m1_rx_bus2
+               &gmac1m1_rgmii_clk
+               &gmac1m1_rgmii_bus
+               &eth1m1_pins
+               &gmac1m1_clkinout>;

tx_delay = <0x4f>;
rx_delay = <0x26>;

phy-handle = <&rgmii_phy1>;
status = "okay";
};

&mdio1 {
    rgmii_phy1: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
+       clocks = <&cru CLK_MAC1_OUT>;
    };
};

```

3.11.5 RGMII Crystal 25M for PHY, PLL output 125M for TX_CLK

- gmac0

```

&gmac0 {
+   phy-mode = "rgmii";
+   clock_in_out = "output";
+   assigned-clocks = <&cru SCLK_GMAC0_RX_TX>, <&cru SCLK_GMAC0>;
+   assigned-clock-parents = <&cru SCLK_GMAC0_RGMII_SPEED>;
+   assigned-clock-rates = <0>, <125000000>;

snps,reset-gpio = <&gpio2 RK_PD3 GPIO_ACTIVE_LOW>;
snps,reset-active-low;
/* Reset time is 20ms, 100ms for rtl8211f */
snps,reset-delays-us = <0 20000 100000>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&gmac0_miim
+               &gmac0_tx_bus2

```

```

+           &gmac0_rx_bus2
+           &gmac0_rgmii_clk
+           &gmac0_rgmii_bus>;

    tx_delay = <0x3c>;
    rx_delay = <0x2f>;
    phy-handle = <&rgmii_phy0>;
    status = "okay";
};

&mdio0 {
    rgmii_phy0: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};

```

- gmac1m0

```

&gmac1 {
+   phy-mode = "rgmii";
+   clock_in_out = "output";

    snps,reset-gpio = <&gpio2 RK_PD1 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

+   assigned-clocks = <&cru SCLK_GMAC1_RX_TX>, <&cru SCLK_GMAC1>;
+   assigned-clock-parents = <&cru SCLK_GMAC1_RGMII_SPEED>;
+   assigned-clock-rates = <0>, <125000000>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&gmac1m0_miim
+               &gmac1m0_tx_bus2
+               &gmac1m0_rx_bus2
+               &gmac1m0_rgmii_clk
+               &gmac1m0_rgmii_bus>;

    tx_delay = <0x4f>;
    rx_delay = <0x26>;

    phy-handle = <&rgmii_phy1>;
    status = "okay";
};

&mdio1 {
    rgmii_phy1: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};

```

- gmac1m1

```
&gmac1 {
+     phy-mode = "rgmii";
+     clock_in_out = "output";

    snps,reset-gpio = <&gpio2 RK_PD1 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

+     assigned-clocks = <&cru SCLK_GMAC1_RX_TX>, <&cru SCLK_GMAC1>;
+     assigned-clock-parents = <&cru SCLK_GMAC1_RGMII_SPEED>;
+     assigned-clock-rates = <0>, <125000000>;

+     pinctrl-names = "default";
+     pinctrl-0 = <&gmac1m1_miim
+                 &gmac1m1_tx_bus2
+                 &gmac1m1_rx_bus2
+                 &gmac1m1_rgmii_clk
+                 &gmac1m1_rgmii_bus>;

    tx_delay = <0x4f>;
    rx_delay = <0x26>;

    phy-handle = <&rgmii_phy1>;
    status = "okay";
};

&mdio1 {
    rgmii_phy1: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};
```

3.11.6 RGMII Crystal 25M for PHY, RGMII_CLK input 125M for TX_CLK

- gmac0

```
&gmac0 {
+     phy-mode = "rgmii";
+     clock_in_out = "input";
+     assigned-clocks = <&cru SCLK_GMAC0_RX_TX>, <&cru SCLK_GMAC0>;
+     assigned-clock-parents = <&cru SCLK_GMAC0_RGMII_SPEED>, <&gmac0_clkin>;
+     assigned-clock-rates = <0>, <125000000>;

    snps,reset-gpio = <&gpio2 RK_PD3 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;
```

```

+       pinctrl-names = "default";
+       pinctrl-0 = <&gmac0_miim
+                   &gmac0_tx_bus2
+                   &gmac0_rx_bus2
+                   &gmac0_rgmii_clk
+                   &gmac0_rgmii_bus
+                   &gmac0_clkinout>;

       tx_delay = <0x3c>;
       rx_delay = <0x2f>;
       phy-handle = <&rgmii_phy0>;
       status = "okay";
};

&mdio0 {
    rgmii_phy0: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};

```

- gmac1m0

```

&gmac1 {
+       phy-mode = "rgmii";
+       clock_in_out = "input";

       snps,reset-gpio = <&gpio2 RK_PD1 GPIO_ACTIVE_LOW>;
       snps,reset-active-low;
       /* Reset time is 20ms, 100ms for rtl8211f */
       snps,reset-delays-us = <0 20000 100000>;

+       assigned-clocks = <&cru SCLK_GMAC1_RX_TX>, <&cru SCLK_GMAC1>;
+       assigned-clock-parents = <&cru SCLK_GMAC1_RGMII_SPEED>, <&gmac1_clkin>;
+       assigned-clock-rates = <0>, <125000000>;

+       pinctrl-names = "default";
+       pinctrl-0 = <&gmac1m0_miim
+                   &gmac1m0_tx_bus2
+                   &gmac1m0_rx_bus2
+                   &gmac1m0_rgmii_clk
+                   &gmac1m0_rgmii_bus
+                   &gmac1m0_clkinout>;

       tx_delay = <0x4f>;
       rx_delay = <0x26>;

       phy-handle = <&rgmii_phy1>;
       status = "okay";
};

&mdio1 {

```

```

rgmii_phy1: phy@0 {
    compatible = "ethernet-phy-ieee802.3-c22";
    reg = <0x0>;
};
};

```

- gmac1m1

```

&gmac1 {
+   phy-mode = "rgmii";
+   clock_in_out = "input";

    snps,reset-gpio = <&gpio2 RK_PD1 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

+   assigned-clocks = <&cru SCLK_GMAC1_RX_TX>, <&cru SCLK_GMAC1>;
+   assigned-clock-parents = <&cru SCLK_GMAC1_RGMII_SPEED>, <&gmac1_clkin>;
+   assigned-clock-rates = <0>, <125000000>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&gmac1m1_miim
+               &gmac1m1_tx_bus2
+               &gmac1m1_rx_bus2
+               &gmac1m1_rgmii_clk
+               &gmac1m1_rgmii_bus
+               &gmac1m1_clkout>;

    tx_delay = <0x4f>;
    rx_delay = <0x26>;

    phy-handle = <&rgmii_phy1>;
    status = "okay";
};

&mdio1 {
    rgmii_phy1: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};

```

3.11.7 SGMII

DTS, in addition to configuring the gmac and mac phy nodes, also requires configuring the xpcs and combophy nodes.

- combophy

The property `rockchip,sgmii-mac-sel` indicates which gmac is being used:

```
&combphy1_usq {
+     rockchip,sgmii-mac-sel = <0>; /* Use gmac0 for sgmii */
    status = "okay";
};
```

- xpcs

```
&xpcs {
    status = "okay";
};
```

- gmac0

```
&gmac0 {
    phy-mode = "sgmii";

    rockchip,pipegrf = <&pipegrf>;
    rockchip,xpcs = <&xpcs>;

    snps,reset-gpio = <&gpio2 RK_PC2 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 20000 100000>;

    assigned-clocks = <&cru SCLK_GMAC0_RX_TX>;
    assigned-clock-parents = <&gmac0_xpcscclk>;

    pinctrl-names = "default";
    pinctrl-0 = <&gmac0_miim>;

    power-domains = <&power RK3568_PD_PIPE>;
    phys = <&combphy1_usq PHY_TYPE_SGMII>;
    phy-handle = <&sgmii_phy>;
    status = "okay";
};

&mdio0 {
    sgmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
    };
};
```

- gmac1

```
&gmac1 {
    phy-mode = "sgmii";

    rockchip,pipegrf = <&pipegrf>;
    rockchip,xpcs = <&xpcs>;

    snps,reset-gpio = <&gpio2 RK_PC2 GPIO_ACTIVE_LOW>;
```

```

snps,reset-active-low;
snps,reset-delays-us = <0 20000 100000>;

assigned-clocks = <&cru SCLK_GMAC1_RX_TX>;
assigned-clock-parents = <&gmac1_xpcscclk>;

pinctrl-names = "default";
pinctrl-0 = <&gmac1_miim>;

power-domains = <&power RK3568_PD_PIPE>;
phys = <&combphy1_usq PHY_TYPE_SGMII>;
phy-handle = <&sgmii_phy>;
status = "okay";
};

&mdio1 {
    sgmii_phy: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
    };
};

```

3.11.8 QSGMII

Similar to SGMII, DTS in addition to configuring the gmac and mac phy nodes, also needs to configure the xpcs and combophy nodes.

- combophy

```

&combphy2_psq {
    status = "okay";
};

```

- xpcs

```

&xpcs {
    status = "okay";
};

```

```

&gmac0 {
    phy-supply = <&pcie20_3v3>;
    phy-mode = "qsgmii";
    rockchip,xpcs = <&xpcs>;

    snps,reset-gpio = <&gpio2 RK_PC2 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 20000 100000>;

    assigned-clocks = <&cru SCLK_GMAC0_RX_TX>;
    assigned-clock-parents = <&gmac0_xpcscclk>;

```

```

pinctrl-names = "default";
pinctrl-0 = <&gmac0_miim>;

power-domains = <&power RK3568_PD_PIPE>;
phys = <&combphy2_psq PHY_TYPE_QSGMII>;
phy-handle = <&qsgmii_phy0>;

status = "okay";
};

&gmac1 {
    phy-supply = <&pcie20_3v3>;
    phy-mode = "qsgmii";

    assigned-clocks = <&cru SCLK_GMAC1_RX_TX>;
    assigned-clock-parents = <&gmac1_xpcscclk>;

    power-domains = <&power RK3568_PD_PIPE>;
    phy-handle = <&qsgmii_phy1>;

    status = "okay";
};

&mdio0 {
    qsgmii_phy0: phy@0 {
        compatible = "ethernet-phy-id001c.c942", "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
    qsgmii_phy1: phy@1 {
        compatible = "ethernet-phy-id001c.c942", "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
    };
    qsgmii_phy2: phy@2 {
        compatible = "ethernet-phy-id001c.c942", "ethernet-phy-ieee802.3-c22";
        reg = <0x2>;
    };
    qsgmii_phy3: phy@3 {
        compatible = "ethernet-phy-id001c.c942", "ethernet-phy-ieee802.3-c22";
        reg = <0x3>;
    };
};

```

3.12 RK3588/RK3576

3.12.1 RK3588/RK3576 RMI Clock Output

- gmac0

```
&gmac0 {
```

```

+     phy-mode = "rmii";
+     clock_in_out = "output";

snps,reset-gpio = <&gpio4 RK_PB3 GPIO_ACTIVE_LOW>;
snps,reset-active-low;
/* Reset time is 20ms, 100ms for rtl8211f */
snps,reset-delays-us = <0 20000 100000>;

+     pinctrl-names = "default";
+     pinctrl-0 = <&gmac0_miim
+                 &gmac0_tx_bus2
+                 &gmac0_rx_bus2
+                 &gmac0_clk_inout>;

phy-handle = <&rmii_phy0>;
status = "okay";
};

&mdio0 {
    rmii_phy0: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
    };
};

```

- gmac1:

```

&gmac1 {
+     phy-mode = "rmii";
+     clock_in_out = "output";

snps,reset-gpio = <&gpio3 RK_PB2 GPIO_ACTIVE_LOW>;
snps,reset-active-low;
/* Reset time is 20ms, 100ms for rtl8211f */
snps,reset-delays-us = <0 20000 100000>;

+     pinctrl-names = "default";
+     pinctrl-0 = <&gmac1_miim
+                 &gmac1_tx_bus2
+                 &gmac1_rx_bus2
+                 &gmac1_clk_inout>;

phy-handle = <&rmii_phy1>;
status = "okay";
};

&mdio1 {
    rmii_phy1: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
    };
};

```

3.12.2 RK3588/RK3576 RMI Clock Input

- gmac0

```
&gmac0 {
+   phy-mode = "rmii";
+   clock_in_out = "input";

   snps,reset-gpio = <&gpio4 RK_PB3 GPIO_ACTIVE_LOW>;
   snps,reset-active-low;
   /* Reset time is 20ms, 100ms for rtl8211f */
   snps,reset-delays-us = <0 20000 100000>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&gmac0_miim
+               &gmac0_tx_bus2
+               &gmac0_rx_bus2
+               &gmac0_clk inout>;

   phy-handle = <&rmii_phy0>;
   status = "okay";
};

&mdio0 {
   rmii_phy0: phy@1 {
       compatible = "ethernet-phy-ieee802.3-c22";
       reg = <0x1>;
   };
};
```

- gmac1:

```
&gmac1 {
+   phy-mode = "rmii";
+   clock_in_out = "input";

   snps,reset-gpio = <&gpio3 RK_PB2 GPIO_ACTIVE_LOW>;
   snps,reset-active-low;
   /* Reset time is 20ms, 100ms for rtl8211f */
   snps,reset-delays-us = <0 20000 100000>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&gmac1_miim
+               &gmac1_tx_bus2
+               &gmac1_rx_bus2
+               &gmac1_clk inout>;

   phy-handle = <&rmii_phy1>;
   status = "okay";
};
```

```

&mdio1 {
    rmii_phy1: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
    };
};

```

3.12.3 RGMII PLL Output 25M for PHY, PLL Output 125M for TX_CLK

- gmac0

```

&gmac0 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    + phy-mode = "rgmii-rxid";
    clock_in_out = "output";

    snps,reset-gpio = <&gpio4 RK_PB3 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    + pinctrl-names = "default";
    + pinctrl-0 = <&gmac0_miim
    +             &gmac0_tx_bus2
    +             &gmac0_rx_bus2
    +             &gmac0_rgmii_clk
    +             &gmac0_rgmii_bus
    +             &eth0_pins>;

    tx_delay = <0x45>;
    /* rx_delay = <0x43>; */

    phy-handle = <&rgmii_phy0>;
    status = "okay";
};

&mdio0 {
    rgmii_phy0: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
        + clocks = <&cru REFCLK025M_ETH0_OUT>;
    };
};

```

- gmac1

```

&gmac1 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    + phy-mode = "rgmii-rxid";
    + clock_in_out = "output";

```

```

snps,reset-gpio = <&gpio3 RK_PB2 GPIO_ACTIVE_LOW>;
snps,reset-active-low;
/* Reset time is 20ms, 100ms for rtl8211f */
snps,reset-delays-us = <0 20000 100000>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&gmac1_miim
+               &gmac1_tx_bus2
+               &gmac1_rx_bus2
+               &gmac1_rgmii_clk
+               &gmac1_rgmii_bus
+               &eth1_pins>;

tx_delay = <0x45>;
/* rx_delay = <0x43>; */

phy-handle = <&rgmii_phy1>;
status = "okay";
};

&mdio1 {
    rgmii_phy1: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
        +   clocks = <&cru REFCLK025M_ETH0_OUT>;
    };
};

```

3.12.4 RGMII PLL Output 25M for PHY, RGMII_CLK Input 125M for TX_CLK

- gmac0

```

&gmac0 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    +   phy-mode = "rgmii-rxid";
    +   clock_in_out = "input";

    snps,reset-gpio = <&gpio4 RK_PB3 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    +   pinctrl-names = "default";
    +   pinctrl-0 = <&gmac0_miim
    +               &gmac0_tx_bus2
    +               &gmac0_rx_bus2
    +               &gmac0_rgmii_clk
    +               &gmac0_rgmii_bus
    +               &gmac0_clk_inout
    +               &eth0_pins>;

```

```

tx_delay = <0x45>;
/* rx_delay = <0x43>; */

phy-handle = <&rgmii_phy0>;
status = "okay";
};

&mdio0 {
    rgmii_phy0: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
        +   clocks = <&cru REFCLK025M_ETH0_OUT>;
    };
};

```

- gmac1

```

&gmac1 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    +   phy-mode = "rgmii-rxid";
    +   clock_in_out = "input";

    snps,reset-gpio = <&gpio3 RK_PB2 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    +   pinctrl-names = "default";
    +   pinctrl-0 = <&gmac1_miim
    +               &gmac1_tx_bus2
    +               &gmac1_rx_bus2
    +               &gmac1_rgmii_clk
    +               &gmac1_rgmii_bus
    +               &gmac1_clkinout
    +               &eth1_pins>;

    tx_delay = <0x45>;
    /* rx_delay = <0x43>; */

    phy-handle = <&rgmii_phy1>;
    status = "okay";
};

&mdio1 {
    rgmii_phy1: phy@1 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x1>;
        +   clocks = <&cru REFCLK025M_ETH0_OUT>;
    };
};

```

3.12.5 RGMII Crystal 25M for PHY, PLL output 125M for TX_CLK

- gmac0

```
&gmac0 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    +   phy-mode = "rgmii-rxid";
    +   clock_in_out = "output";

    snps,reset-gpio = <&gpio4 RK_PB3 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    +   pinctrl-names = "default";
    +   pinctrl-0 = <&gmac0_miim
    +               &gmac0_tx_bus2
    +               &gmac0_rx_bus2
    +               &gmac0_rgmii_clk
    +               &gmac0_rgmii_bus>;

    tx_delay = <0x45>;
    /* rx_delay = <0x43>; */

    phy-handle = <&rgmii_phy0>;
    status = "okay";
};

&mdio0 {
    rgmii_phy0: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};
```

- gmac1

```
&gmac1 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    +   phy-mode = "rgmii-rxid";
    +   clock_in_out = "output";

    snps,reset-gpio = <&gpio3 RK_PB2 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    +   pinctrl-names = "default";
    +   pinctrl-0 = <&gmac1_miim
    +               &gmac1_tx_bus2
    +               &gmac1_rx_bus2
```

```

+           &gmac1_rgmii_clk
+           &gmac1_rgmii_bus>;

tx_delay = <0x45>;
/* rx_delay = <0x43>; */

phy-handle = <&rgmii_phy1>;
status = "okay";
};

&mdio1 {
    rgmii_phy1: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};

```

3.12.6 RGMII Crystal 25M for PHY, RGMII_CLK input 125M for TX_CLK

- gmac0

```

&gmac0 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
+   phy-mode = "rgmii-rxid";
+   clock_in_out = "input";

    snps,reset-gpio = <&gpio4 RK_PB3 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&gmac0_miim
+               &gmac0_tx_bus2
+               &gmac0_rx_bus2
+               &gmac0_rgmii_clk
+               &gmac0_rgmii_bus
+               &gmac0_clk inout>;

    tx_delay = <0x45>;
    /* rx_delay = <0x43>; */

    phy-handle = <&rgmii_phy0>;
    status = "okay";
};

&mdio0 {
    rgmii_phy0: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};

```

```
};
```

- gmac1

```
&gmac1 {
    /* Use rgmii-rxid mode to disable rx delay inside Soc */
    +   phy-mode = "rgmii-rxid";
    +   clock_in_out = "input";

    snps,reset-gpio = <&gpio3 RK_PB2 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

    +   pinctrl-names = "default";
    +   pinctrl-0 = <&gmac1_miim
    +               &gmac1_tx_bus2
    +               &gmac1_rx_bus2
    +               &gmac1_rgmii_clk
    +               &gmac1_rgmii_bus
    +               &gmac1_clkinout>;

    tx_delay = <0x45>;
    /* rx_delay = <0x43>; */

    phy-handle = <&rgmii_phy1>;
    status = "okay";
};

&mdio1 {
    rgmii_phy1: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};
```

3.13 RV1108

3.13.1 RV1108 RMI Clock Input

```
+gmac_clkin: gmac_clkin {
+   compatible = "fixed-clock";
+   clock-output-names = "gmac_clkin";
+   clock-frequency = <50000000>;
+   #clock-cells = <0>;
+};

&gmac {
+   phy-mode = "rmii";
```

```

+     clock_in_out = "input";
+     assigned-clocks = <&cru SCLK_MAC>;
+     assigned-clock-parents = <&gmac_clkin>;
snps,reset-gpio = <&gpio3 12 0>;
snps,reset-active-low;
snps,reset-delays-us = <0 20000 100000>;
+     pinctrl-names = "default";
+     pinctrl-0 = <&rmii_pins>;
status = "ok";
};

```

3.13.2 RV1108 RMI Clock Output

```

&gmac {
+     phy-mode = "rmii";
+     clock_in_out = "output";
+     assigned-clocks = <&cru SCLK_MAC>;
+     assigned-clock-rates = <50000000>;
snps,reset-gpio = <&gpio3 12 0>;
snps,reset-active-low;
snps,reset-delays-us = <0 20000 100000>;
+     pinctrl-names = "default";
+     pinctrl-0 = <&rmii_pins>;
status = "ok";
};

```

3.14 RV1126

3.14.1 RGMII PLL Output 25M for PHY, PLL Output 125M for TX_CLK

- gmac m0

```

&gmac {
+     phy-mode = "rgmii";
+     clock_in_out = "output";

snps,reset-gpio = <&gpio3 RK_PA0 GPIO_ACTIVE_LOW>;
snps,reset-active-low;
/* Reset time is 20ms, 100ms for rtl8211f */
snps,reset-delays-us = <0 20000 100000>;

+     assigned-clocks = <&cru CLK_GMAC_SRC>, <&cru CLK_GMAC_TX_RX>, <&cru
CLK_GMAC_ETHERNET_OUT>;
+     assigned-clock-parents = <&cru CLK_GMAC_SRC_M0>, <&cru RGMII_MODE_CLK>;
+     assigned-clock-rates = <125000000>, <0>, <25000000>;

+     pinctrl-names = "default";
+     pinctrl-0 = <&rgmiim0_miim &rgmiim0_bus2 &rgmiim0_bus4 &clkm0_out_ethernet>;

```

```

tx_delay = <0x2a>;
rx_delay = <0x1a>;

phy-handle = <&phy>;
status = "okay";
};

&mdio {
    phy: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
+        clocks = <&cru CLK_GMAC_ETHERNET_OUT>;
    };
};

```

- gmac m1

```

&gmac {
+    phy-mode = "rgmii";
+    clock_in_out = "output";

    snps,reset-gpio = <&gpio3 RK_PA0 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    /* Reset time is 20ms, 100ms for rtl8211f */
    snps,reset-delays-us = <0 20000 100000>;

+    assigned-clocks = <&cru CLK_GMAC_SRC>, <&cru CLK_GMAC_TX_RX>, <&cru
CLK_GMAC_ETHERNET_OUT>;
+    assigned-clock-parents = <&cru CLK_GMAC_SRC_M1>, <&cru RGMII_MODE_CLK>;
+    assigned-clock-rates = <125000000>, <0>, <25000000>;

+    pinctrl-names = "default";
+    pinctrl-0 = <&rgmiim1_miim &rgmiim1_bus2 &rgmiim1_bus4 &clkm1_out_ethernet>;

    tx_delay = <0x2a>;
    rx_delay = <0x1a>;

    phy-handle = <&phy>;
    status = "okay";
};

&mdio {
    phy: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
+        clocks = <&cru CLK_GMAC_ETHERNET_OUT>;
    };
};

```

3.14.2 RGMII PLL Output 25M for PHY, RGMII Clock Input 125M for TX_CLK

- gmac m0

```
&gmac {
+   phy-mode = "rgmii";
+   clock_in_out = "input";

   snps,reset-gpio = <&gpio3 RK_PA0 GPIO_ACTIVE_LOW>;
   snps,reset-active-low;
   /* Reset time is 20ms, 100ms for rtl8211f */
   snps,reset-delays-us = <0 20000 100000>;

+   assigned-clocks = <&cru CLK_GMAC_SRC>, <&cru CLK_GMAC_TX_RX>, <&cru
CLK_GMAC_ETHERNET_OUT>;
+   assigned-clock-parents = <&cru CLK_GMAC_SRC_M0>, <&cru RGMII_MODE_CLK>;
+   assigned-clock-rates = <125000000>, <0>, <25000000>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&rgmiim0_miim &rgmiim0_bus2 &rgmiim0_bus4 &clkm0_out_ethernet>;

   tx_delay = <0x2a>;
   rx_delay = <0x1a>;

   phy-handle = <&phy>;
   status = "okay";
};

&mdio {
   phy: phy@0 {
       compatible = "ethernet-phy-ieee802.3-c22";
       reg = <0x0>;
+       clocks = <&cru CLK_GMAC_ETHERNET_OUT>;
   };
};
```

- gmac m1

```
&gmac {
+   phy-mode = "rgmii";
+   clock_in_out = "input";

   snps,reset-gpio = <&gpio3 RK_PA0 GPIO_ACTIVE_LOW>;
   snps,reset-active-low;
   /* Reset time is 20ms, 100ms for rtl8211f */
   snps,reset-delays-us = <0 20000 100000>;

+   assigned-clocks = <&cru CLK_GMAC_SRC>, <&cru CLK_GMAC_TX_RX>, <&cru
CLK_GMAC_ETHERNET_OUT>;
+   assigned-clock-parents = <&cru CLK_GMAC_SRC_M1>, <&cru RGMII_MODE_CLK>;
+   assigned-clock-rates = <125000000>, <0>, <25000000>;
```

```

+   pinctrl-names = "default";
+   pinctrl-0 = <&rgmiim1_miim &rgmiim1_bus2 &rgmiim1_bus4 &clkm1_out_ethernet>;

   tx_delay = <0x2a>;
   rx_delay = <0x1a>;

   phy-handle = <&phy>;
   status = "okay";
};

&mdio {
    phy: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
+       clocks = <&cru CLK_GMAC_ETHERNET_OUT>;
    };
};

```

3.14.3 RGMII Crystal 25M for PHY, PLL Output 125M for TX_CLK

- gmac m0

```

&gmac {
+   phy-mode = "rgmii";
+   clock_in_out = "output";

   snps,reset-gpio = <&gpio3 RK_PA0 GPIO_ACTIVE_LOW>;
   snps,reset-active-low;
   /* Reset time is 20ms, 100ms for rtl8211f */
   snps,reset-delays-us = <0 20000 100000>;

+   assigned-clocks = <&cru CLK_GMAC_SRC>, <&cru CLK_GMAC_TX_RX>;
+   assigned-clock-parents = <&cru CLK_GMAC_SRC_M0>, <&cru RGMII_MODE_CLK>;
+   assigned-clock-rates = <125000000>, <0>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&rgmiim0_miim &rgmiim0_bus2 &rgmiim0_bus4 &clkm0_out_ethernet>;

   tx_delay = <0x2a>;
   rx_delay = <0x1a>;

   phy-handle = <&phy>;
   status = "okay";
};

&mdio {
    phy: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};

```

```
};
```

- gmac m1

```
&gmac {
+   phy-mode = "rgmii";
+   clock_in_out = "output";

   snps,reset-gpio = <&gpio3 RK_PA0 GPIO_ACTIVE_LOW>;
   snps,reset-active-low;
   /* Reset time is 20ms, 100ms for rtl8211f */
   snps,reset-delays-us = <0 20000 100000>;

+   assigned-clocks = <&cru CLK_GMAC_SRC>, <&cru CLK_GMAC_TX_RX>;
+   assigned-clock-parents = <&cru CLK_GMAC_SRC_M1>, <&cru RGMII_MODE_CLK>;
+   assigned-clock-rates = <125000000>, <0>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&rgmiim1_miim &rgmiim1_bus2 &rgmiim1_bus4 &clkm1_out_ethernet>;

   tx_delay = <0x2a>;
   rx_delay = <0x1a>;

   phy-handle = <&phy>;
   status = "okay";
};

&mdio {
   phy: phy@0 {
       compatible = "ethernet-phy-ieee802.3-c22";
       reg = <0x0>;
   };
};
```

3.14.4 RGMII Crystal 25M for PHY, RGMII_CLK input 125M for TX_CLK

- gmac m0

```
&gmac {
+   phy-mode = "rgmii";
+   clock_in_out = "input";

   snps,reset-gpio = <&gpio3 RK_PA0 GPIO_ACTIVE_LOW>;
   snps,reset-active-low;
   /* Reset time is 20ms, 100ms for rtl8211f */
   snps,reset-delays-us = <0 20000 100000>;

+   assigned-clocks = <&cru CLK_GMAC_RGMII_M0>, <&cru CLK_GMAC_SRC_M0>, <&cru CLK_GMAC_SRC>,
+   <&cru CLK_GMAC_TX_RX>;
```

```

+ assigned-clock-parents = <&gmac_clkin_m0>, <&cru CLK_GMAC_RGMII_M0>, <&cru
CLK_GMAC_SRC_M0>, <&cru RGMII_MODE_CLK>;

+ pinctrl-names = "default";
+ pinctrl-0 = <&rgmiim0_miim &rgmiim0_bus2 &rgmiim0_bus4>;

tx_delay = <0x2a>;
rx_delay = <0x1a>;

phy-handle = <&phy>;
status = "okay";
};

&mdio {
    phy: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};

```

- gmac m1

```

&gmac {
+ phy-mode = "rgmii";
+ clock_in_out = "input";

snps,reset-gpio = <&gpio3 RK_PA0 GPIO_ACTIVE_LOW>;
snps,reset-active-low;
/* Reset time is 20ms, 100ms for rtl8211f */
snps,reset-delays-us = <0 20000 100000>;

+ assigned-clocks = <&cru CLK_GMAC_SRC>, <&cru CLK_GMAC_TX_RX>, <&cru
CLK_GMAC_ETHERNET_OUT>;
+ assigned-clock-parents = <&cru CLK_GMAC_SRC_M1>, <&cru RGMII_MODE_CLK>;
+ assigned-clock-rates = <125000000>, <0>, <250000000>;

+ pinctrl-names = "default";
+ pinctrl-0 = <&rgmiim1_miim &rgmiim1_bus2 &rgmiim1_bus4>;

tx_delay = <0x2a>;
rx_delay = <0x1a>;

phy-handle = <&phy>;
status = "okay";
};

&mdio {
    phy: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};

```

3.14.5 RV1126 RMI Clock Output

- gmac m0

```
&gmac {
+   phy-mode = "rmii";
+   clock_in_out = "output";

   snps,reset-gpio = <&gpio3 RK_PC5 GPIO_ACTIVE_LOW>;
   snps,reset-active-low;
   snps,reset-delays-us = <0 50000 50000>;

+   assigned-clocks = <&cru CLK_GMAC_SRC_M0>, <&cru CLK_GMAC_SRC>, <&cru CLK_GMAC_TX_RX>;
+   assigned-clock-rates = <0>, <50000000>;
+   assigned-clock-parents = <&cru CLK_GMAC_RGMII_M0>, <&cru CLK_GMAC_SRC_M0>, <&cru
RMI_MODE_CLK>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&rmii0_miim &rgmii0_rxer &rmii0_bus2 &rgmii0_mclkinout>;

   phy-handle = <&phy>;
   status = "okay";
};

&mdio {
   phy: phy@0 {
       compatible = "ethernet-phy-ieee802.3-c22";
       reg = <0x0>;
   };
};
```

- gmac m1

```
&gmac {
+   phy-mode = "rmii";
+   clock_in_out = "output";

   snps,reset-gpio = <&gpio3 RK_PC5 GPIO_ACTIVE_LOW>;
   snps,reset-active-low;
   snps,reset-delays-us = <0 50000 50000>;

+   assigned-clocks = <&cru CLK_GMAC_SRC_M1>, <&cru CLK_GMAC_SRC>, <&cru CLK_GMAC_TX_RX>;
+   assigned-clock-rates = <0>, <50000000>;
+   assigned-clock-parents = <&cru CLK_GMAC_RGMII_M1>, <&cru CLK_GMAC_SRC_M1>, <&cru
RMI_MODE_CLK>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&rmii1_miim &rgmii1_rxer &rmii10_bus2 &rgmii1_mclkinout>;

   phy-handle = <&phy>;
   status = "okay";
```

```
};

&mdio {
    phy: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};
```

3.14.6 RV1126 RMI Clock Input

- gmac m0

```
+&gmac_clkin_m0 {
+    clock-frequency = <50000000>;
+};

&gmac {
+    phy-mode = "rmii";
+    clock_in_out = "input";

    snps,reset-gpio = <&gpio3 RK_PC5 GPIO_ACTIVE_LOW>;
    snps,reset-active-low;
    snps,reset-delays-us = <0 50000 50000>;

+    assigned-clocks = <&cru CLK_GMAC_RGMII_M0>, <&cru CLK_GMAC_SRC_M0>, <&cru CLK_GMAC_SRC>,
<&cru CLK_GMAC_TX_RX>;
+    assigned-clock-rates = <0>, <0>, <50000000>;
+    assigned-clock-parents = <&gmac_clkin_m0>, <&cru CLK_GMAC_RGMII_M0>, <&cru
CLK_GMAC_SRC_M0>, <&cru RMII_MODE_CLK>;

+    pinctrl-names = "default";
+    pinctrl-0 = <&rmii0_miim &rgmii0_rxer &rmii0_bus2 &rgmii0_mclkinout_level0>;

    phy-handle = <&phy>;
    status = "okay";
};

&mdio {
    phy: phy@0 {
        compatible = "ethernet-phy-ieee802.3-c22";
        reg = <0x0>;
    };
};
```

- gmac m1

```
+&gmac_clkin_m1 {
+    clock-frequency = <50000000>;
+};
```

```

&gmac {
+   phy-mode = "rmii";
+   clock_in_out = "input";

   snps,reset-gpio = <&gpio3 RK_PC5 GPIO_ACTIVE_LOW>;
   snps,reset-active-low;
   snps,reset-delays-us = <0 50000 50000>;

+   assigned-clocks = <&cru CLK_GMAC_RGMII_M1>, <&cru CLK_GMAC_SRC_M1>, <&cru CLK_GMAC_SRC>,
<&cru CLK_GMAC_TX_RX>;
+   assigned-clock-rates = <0>, <0>, <50000000>;
+   assigned-clock-parents = <&gmac_clkin_m1>, <&cru CLK_GMAC_RGMII_M1>, <&cru
CLK_GMAC_SRC_M1>, <&cru RMII_MODE_CLK>;

+   pinctrl-names = "default";
+   pinctrl-0 = <&rmii1_miim &rgmii1_rxer &rmii1_bus2 &rgmii1_mclkinout_level0>;

   phy-handle = <&phy>;
   status = "okay";
};

&mdio {
   phy: phy@0 {
       compatible = "ethernet-phy-ieee802.3-c22";
       reg = <0x0>;
   };
};

```